

Table 3: Generative performance. The optimal and suboptimal results are highlighted in red and blue respectively. M and N represent the number of parts.

Category	Model	JSD ↓	MMD ↓		COV %, ↑	
			CD	EMD	CD	EMD
Chair	TreeGAN	0.119	0.0016	0.101	58	30
	MRGAN	0.246	0.0021	0.166	67	23
	EditVAE (M=7)	0.063	0.0014	0.082	46	32
	EditVAE (M=3)	0.031	0.0017	0.101	45	39
	SGAS (N=7)	0.047	0.0020	0.076	60	58
Airplane	TreeGAN	0.097	0.0004	0.068	61	20
	MRGAN	0.243	0.0006	0.114	75	21
	EditVAE (M=6)	0.043	0.0004	0.024	39	30
	EditVAE (M=3)	0.044	0.0005	0.067	23	17
	SGAS (N=6)	0.036	0.0004	0.039	61	58
Table	TreeGAN	0.077	0.0018	0.082	71	48
	MRGAN	0.287	0.0020	0.155	78	31
	EditVAE (M=5)	0.081	0.0016	0.071	42	27
	EditVAE (M=3)	0.042	0.0017	0.130	39	30
	SGAS (N=5)	0.057	0.0020	0.069	65	65

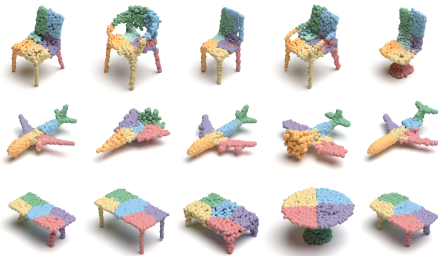


Figure 7: Point clouds generated by SGAS, colored by parts.

Ground Truth part datasets; (b) training SGAS on these part datasets. Specifically, we first modify the multi-branch AX-form to output one structure point per branch. Second, we use these structure points to co-segment the Ground Truth point clouds into n part datasets. Third, the segmented parts are pre-encoded into latent features. Finally, these latent features are used to supervise the training of SGAS. However, we found that there might be some large gaps between parts during generation. Therefore, to achieve seamless generation, during unsupervised part segmentation, we further expand the number of points per part from $\lfloor \frac{2048}{n} \rfloor$ to $(1 + \gamma) \lfloor \frac{2048}{n} \rfloor$. Here $\gamma = 0.1$. In the final output, we down-sample the point cloud to 2048 points.

The quantitative results are shown in Table 3. The metrics are proposed by ?, and the results of previous methods are obtained from EditVAE (?). MMD and COV represent the quality and diversity of the generated results respectively. It can be found that SGAS achieves excellent results overall, with the most number of top two metrics. Especially on COV-EMD, which represents diversity, SGAS has a significant improvement. Figure 7 gives visualized results of the generated point clouds. Different colors correspond to different parts. It intuitively illustrates the diversity and quality of the results generated by SGAS. We also conduct an ablation study on the number of parts in the Airplane category.

Table 4: Ablation results for the number of parts. The optimal and suboptimal results are highlighted in red and blue respectively. $n = 6$ is a suitable number of parts.

#Parts (n)	JSD ↓	MMD ↓		COV %, ↑	
		CD	EMD	CD	EMD
2	0.042	0.0004	0.045	61	47
3	0.039	0.0004	0.043	60	52
5	0.040	0.0005	0.042	60	50
6	0.036	0.0004	0.039	61	58
8	0.040	0.0005	0.044	60	46
13	0.035	0.0005	0.040	60	53

As shown in Table 4, more or fewer parts are not necessarily beneficial to the results. Therefore, we chose the number of parts $N = 6$.

Conclusion

Previous methods do not disentangle each part completely or lack constraints between parts, which leads to poor diversity, fidelity, and quality when performing part editing. In this work, to solve these problems, we first propose a novel four-stage point cloud part editing process. Then based on this process and two new strategies: feature disentanglement and constraint, we propose a part editing model SGAS. It can realize various part editing operations. By introducing metrics from multimodal completion and proposing a new metric TMDS, we establish quantitative evaluations for diversity part editing. In addition, SGAS can be pruned to realize unsupervised part-aware point cloud generation. Experiments show that it performs better than previous methods.

Limitation Since we do not design part offset networks for the generated parts but instead utilize the relatively fixed spatial positions of each generated part to ensure good assembly, SGAS can only achieve part editing for objects with relatively consistent prototypes. For example, SGAS cannot handle part editing for both ceiling lamps and table lamps simultaneously as the spatial order of the parts is opposite. In addition, we also find that the performance of SGAS is limited by the pre-trained autoencoders. The embedded features are better when the parts are normalized. Therefore, it will be beneficial to first generate normalized parts and then design part offset networks to align them in the future.

Acknowledgments

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